

Assignment 03

Part 1: Implement a Spotify Playlist

In part 1, you will implement a music playlist structure capable of adding, removing, and iterating through songs/tracks.

Part 1.1: Implementing a Music Track

The first part of this task will involve implementing a C# *public* class to hold a single music *Track*. This will be a short class but could be expanded on later if needed. The music *Track* class must be capable of holding a *string name*, *string artistName*, *string albumName*, *int durationSec*.

When developing this class, all string entries should be non-null and non-empty, and the duration of the song must be at least 1 second long. If an invalid entry is provided in the setters, throw an *ArgumentException* or *NullArgumentException* with a message which indicates which argument(s) were invalid.

In terms of constructors, you should create a single constructor that takes has the following signature: *string name*, *string artistName*, *string albumName*, *int durationSec*. As before if any are of these values are invalid, throw the appropriate exception. **HINT:** This can easily be done by calling the setters you developed prior.

Part 1.2: Implementing a DoublyLinkedList

The second part of part 1 is to implement a *DoublyLinkedList*. We will be do this ourselves and **not** be using C#'s built-in *List* structure. We will later use our *DoublyLinkedList* to hold our *Playlist*.

An initial *DNode* class has been given to you. This class is used to represent a single node in a *DoublyLinkedList*. Our *DNode* contains a *Track* song, as well as pointers to the previous and next node.

Our *DoublyLinkedList* will have a constructor that takes no arguments but sets the head and tail pointers to null. It also sets the *int* instance variable *size* to 0 as the initial size of the list is 0 when the object is created. I have implemented this constructor to help you get started.

Implement the remaining functions of *DoublyLinkedList* based on the comments in the provided *Part1.cs* file.

Part 1.3: Implementing Playlist

Your *Playlist* class will hold a string instance variables name and count which represents the name of the *Playlist* and the number of songs it contains. You will also need to include an *DoublyLinkedList* instance variable called tracks which holds a collection of *Tracks*.

You will have a single constructor class has a signature: *string name*. Tracks will be added via the Add() method. You will also be able to Delete() tracks from the playlist.

Finally, you will override a ToString() method to display the current Track's details. Additionally, you will have a Next() and sPrevious() method to move forward or backwards through the playlist.

OPTIONAL: You may challenge yourself to add a Shuffle() method. This is a tricky method to implement as it should randomly rearrange the songs in your playlist. There are a few different approaches to this. Later in the course we will learn about interfaces and particularly IList<T> which will make implementing such shuffling easier. Again, this method is optional.